

CS242 (System Software Lab): Assignment III

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Today is : 27th August, 2024

Problem 1 :

I. Install sccs by `sudo apt-get install cssc` (For Ubuntu)

II. Try these commands one after another:

```
$mkdir sccs test
```

```
$cd sccs test
```

```
$echo 'hallo' > quit.c
```

```
$mkdir SCCS
```

```
$sccs create quit.c
```

```
$sccs edit quit.c
```

```
$echo 'bye' >> quit.c
```

```
$sccs delta quit.c
```

The last command asks for a comment input. Then try out these questions.

- 1) Create a file named “iitg.txt” with five different versions and each version has some changes from its previous version.
- 2) How to add a new comment to a 1.3 version of the iitg.txt file ?
- 3) Display the changes made to the 4th version of the iitg.txt file.
- 4) How to check the changes and then retrieves a read-only copy of the new version iitg.txt file.
- 5) How to retrieve the latest version of the file iitg.txt, which was created before 28th Aug 2024 at 2.30pm (or similar date and timing).
- 6) How to exclude delta changes made at the versions 1.2 and 1.4 should not be reflected on the current version of the iitg.txt file

Problem 2 :

Install git in your system. Initialize a git repository either on your own machine with one file (say hello.c containing a hello world program) inside it. Once the repository is set up do the commit operation. Report the commands you used for the initialize and commit purpose.

Problem 3 :

Clone a repository to local directory for modification. Report the command. Make some random changes in the hello world program and check the status of repository. Report the differences from the previous status.

Problem 4 :

Make two copies of the binary a.out file and add these files in the git repository and perform the commit. Remove one a.out file normally (using rm command) and check git status and remove another a.out file with git rm command and check git status. Report the differences in both the git status.

Problem 5 :

Write a command to undo the git repository to first commit using commit id.

Problem 6 :

- Install gdb and ddd in your system.
- Download gdbtest1.cpp from course website. Compile the program for debugging
- Run the program on debugger
- Find out where, you got the issue in the program
- Again : Run the above program on debugger.
- Put the break point at line where main function starts. Put another break point at line number 17 (inside the function divint)
- Step through the lines and insides the function
- At each step print the value of x, y (or a, b)
- Again : Run the above program on debugger.
- Put the break point at line where main function starts. Put another break point at line number 17 (inside the function divint)
- run the program and go upto break point, again continue running program upto next break points.
- At each step print the value of x, y (or a, b)

Problem 7 :

- Download gdbtest2.cpp and gdbtest3.cpp from course website. Compile the program for debugging
- Run the program on debugger
- Find out location of issue using debugger
- find out cause of issue at the locations
- Use backtrace command and
- Print the value of ip (other variable) just before segfault using debug mode

Problem 8 :

- Download gdbtest5.cpp from course website.
- Compile the program for normal mode of execution, run the code. Find out the logical issue
- Compile the program for debugging
- Run the program on debugger, Put the break point at line where main function starts
- Step through the lines using n commands
- Find out (logical error) location of issue using debugger

Problem 9 :

Repeat all the same debugging process for problem 6, 7 and 8 using DDD. Identify the button for breakpoint, run command, next step (s), next step (s), how to show/print variable value.